

Binaural Explorer: Theta. A Csound & Cabbage-based System for Real-Time Auditory Entrainment

Yuqi Wu¹ and Richard Boulanger²

¹ University of California, Berkeley

² Berklee College of Music

yuqiwuyuqi@gmail.com, rboulanger@berklee.edu

Abstract. This paper presents *Binaural Explorer: Theta*, a cross-platform Cabbage plugin that embeds precise theta-frequency binaural beats directly into existing music, generative soundscapes, and live DAW environments. The plugin uses the Csound synthesis engine to apply a real-time spectral phase vocoder, superimposing a neurologically-active theta differential (4.0–7.2 Hz) onto arbitrary audio signals without degrading the underlying sonic structure. Users can substitute source tracks, activate a fully binauralized generative composition mode, and fine-tune target beat frequencies via granular controls. The system includes dynamic visual feedback (a pulsing “beating cabbage” graphic scaled to the live theta rate), an interactive educational mode, and a linked informational pane for psychoacoustic study. A small early-adopter cohort reported reduced sleep latency, enhanced study focus, and a striking expansion of the perceived soundstage. This readily-customizable, open-source tool functions as both a research utility and a pedagogical toolkit, supporting informal clinical research, interactive sonification, and meditation practice.

Keywords: Csound, Cabbage, Binaural Beats, Auditory Entrainment, Audio Plugins, Generative Music, Psychoacoustics

1 Introduction

1.1 Personal Motivation and Problem Context

This project grew out of personal conversations between the authors about chronic sleep latency, a challenge made harder by a twelve-hour continental time difference. While exploring non-pharmacological approaches, they discovered the binaural beat literature but found existing recordings dominated by repetitive sine waves that quickly induced auditory fatigue. A natural question followed: could therapeutic phase relationships be embedded directly into rich musical experiences, rather than presented as isolated clinical stimuli? That question drove the development of *Binaural Explorer: Theta* (Fig. 1), the first, single-technique instrument in what we now intend as a longer-running line of work.

1.2 Bridging Music Technology and Neuroscience, and Teaching Through It

This project treats music technology as a programmable interface for psychoacoustic research and digital health, blurring the line between clinical utility and artistic expression. One persistent challenge in entrainment research is separating physiological entrainment from placebo effects [8]. Rather than resolving that debate, *Binaural Explorer: Theta* is designed to serve as an open-source, reproducible platform for future double-blind studies comparing entrainment-enabled audio against identical baseline material.

Early test participants consistently preferred stimuli embedded in rich musical material over pure sine-wave tracks, describing binauralized versions as more spacious and three-dimensional while reporting that melody, harmony, and timbre remained fully intact. A dedicated information pane covers theta wave properties, interaural phase difference mechanics, and safety instructions, with active hyperlinks to open-access clinical data repositories and Csound documentation. The plugin is as much a teaching object as a research tool: several of the observations below, including the one that prompted Section 2.3, came from students and colleagues using the interface, not from the authors alone.

2 System Architecture and DSP Implementation

Binaural Explorer is built with the Cabbage framework [2], which compiles unified Csound documents (.csd) into native VST, VST3, and AU plugins and standalone apps, enabling deployment in Reaper, Ableton Live, and Logic Pro.

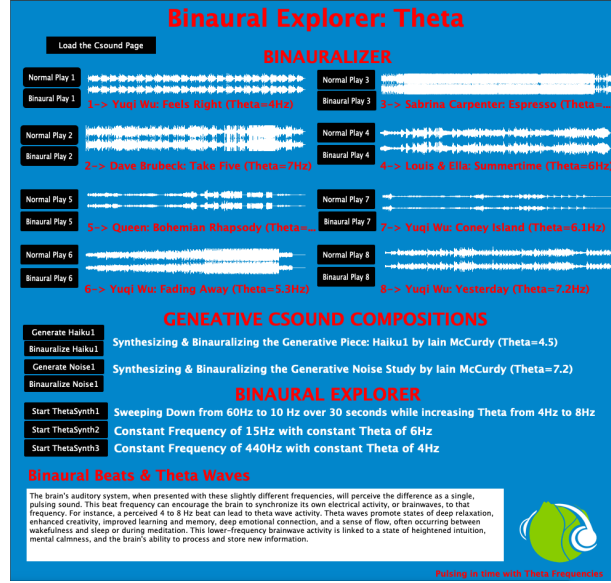


Fig. 1. *Binaural Explorer: Theta* main user interface: Binauralizer section (eight preloaded tracks with normal/binaural A–B playback), Generative Csound Compositions, Binaural Explorer synth modes, and the educational Binaural Beats & Theta Waves information pane.

2.1 High-Level Architecture

The plugin establishes a dual-layer architecture. The **Front-End (GUI Layer)** is a Cabbage-native interface exposing carrier frequency f_c , beat frequency f_b , spatial depth, and mode selection; GUI parameters are polled by `cabbageGetValue` in an always-on `instr 1`, which dispatches playback instruments (a play-button channel, for instance, starts or stops a `SamplePlayback1` event on a single conditional). The **Back-End (DSP Engine)** consists of Csound audio-rate (a -rate) and control-rate (k -rate) loops that generate, modulate, and spatialize the binaural beat signal with microsecond precision. Fig. 2 shows the complete signal flow from GUI to stereo output. The version described here implements one entrainment mechanism at a time; Section 4 outlines a successor that lets several run side by side.

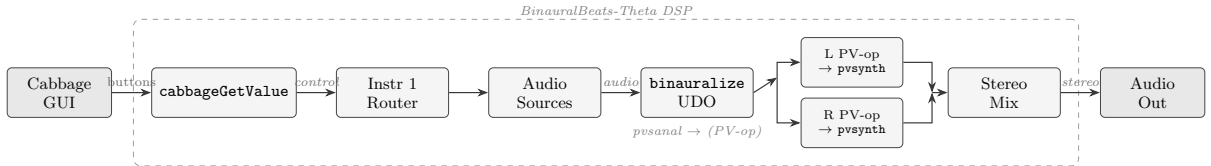


Fig. 2. Functional block diagram of the Binaural Explorer (*theta*) Cabbage plugin, showing GUI-to-DSP control routing via `cabbageGetValue` and the `binauralize` spectral pitch-splitting path to stereo output. “PV-op” is `pvscale` in the original v1 UDO and `pvsynth` in the corrected UDO of Section 2.3; the surrounding topology is unchanged.

GUI parameters include: **Carrier Frequency** (f_c , typically 200–2000 Hz), setting the audible pitch center; **Beat Frequency** (f_b , 1–40 Hz), determining the neural entrainment target (4–8 Hz theta for sleep onset); **Spatial Depth**, mapped to HRTF filtering opcodes; and **Mode Selection** (Pure Sine Wave, Ambient Texture, or Algorithmic).

2.2 Core DSP Implementation

To preserve melody, harmony, and timbre without time-domain artifacts, the DSP backend uses a Short-Time Fourier Transform (STFT) phase vocoder with a 2048-sample FFT window and $4\times$ overlap [3]. The required phase deviation per hop is:

$$\Delta\phi = 2\pi \cdot f_b \cdot \frac{H}{F_s} \quad (1)$$

where $H = 512$ samples is the hop size and F_s is the system sampling rate (44.1 or 48 kHz). Because manipulation occurs entirely in the frequency domain before inverse-STFT reconstruction, transient smearing is mitigated.

The v1 release built this stage on Dave Seidel’s **binauralize** User Defined Opcode (UDO) [9]. Rather than generating raw sine waves, the UDO tracks spectral centroid via **pvscent** and applies complementary **pvscale** ratios to the left and right channels, preserving harmonic richness while nominally embedding the theta differential.

2.3 A Scaling Artifact in binauralize, and a pvshift Fix

A reviewer of the v1 plugin raised a specific and, on inspection, correct objection: **pvscale** multiplies every analysis bin’s frequency by a common ratio ($\text{krat1} = (\text{kcent} + \text{kdif}) / \text{kcent}$, derived from the tracked centroid **kcent** and target difference **kdif**), so the resulting Hz offset is not constant across the spectrum. A bin at 100 Hz scaled by 1.05 shifts by 5 Hz, the intended theta differential; a bin at 1000 Hz scaled by the same 1.05 shifts by 50 Hz. A binaural beat is defined by an absolute frequency difference, not a ratio, so the upper partials of anything but a narrow, low, single-pitch source drift out of the theta band, and the beat a listener hears depends on a track’s spectral content rather than the f_b set on the GUI. **pvscale** is the right opcode for pitch transposition; it is the wrong one for a fixed-differential binaural beat.

The fix is to replace the multiplicative step with **pvshift**, which adds a signed Hz offset to every analysis bin regardless of that bin’s frequency. Because the offset is already in Hz, centroid tracking is no longer needed: **binauralizeShift** shifts the whole spectrum by $\pm k_{\text{dif}} / 2$ Hz directly. Fig. 3 lists the two UDOS side by side.

v1 — binauralize (pvscale)	Revised — binauralizeShift (pvshift)
<pre>opcode binauralize, aa, ak ain, kdif xin fsig pvsanal ain, 2048, 512, 2048, 1 kcent pvscent fsig if (kcent>0)&&(kdif>0) then kdif /= 2 krat1 = (kcent+kdif)/kcent krat2 = (kcent-kdif)/kcent else krat1 = 1 krat2 = 1 endif fbinL pvscale fsig, krat1, 0 fbinR pvscale fsig, krat2, 0 aoutL pvsynth fbinL aoutR pvsynth fbinR xout aoutL, aoutR endop</pre>	<pre>opcode binauralizeShift, aa, ak ain, kdif xin fsig pvsanal ain, 2048, 512, 2048, 1 kshift = kdif / 2 fbinL pvshift fsig, kshift, 0 fbinR pvshift fsig, -kshift, 0 aoutL pvsynth fbinL aoutR pvsynth fbinR xout aoutL, aoutR endop</pre>

Fig. 3. The original **binauralize** UDO (left, after Seidel [9]), which scales bin frequencies by a centroid-relative ratio and so does not hold a fixed Hz differential across the spectrum, beside its replacement **binauralizeShift** (right), which shifts every bin by a fixed $\pm k_{\text{dif}} / 2$ Hz and needs no centroid tracking.

The correction is not free: **pvscale** preserves harmonic ratios between partials, which is why it never audibly detunes a tonal source but also why its beat is inconsistent. **pvshift** trades that away: a shifted overtone series is no longer an integer multiple of its fundamental, which produces a faint ring-modulation-like coloration. In listening tests this was audible but mild against a carrier above 150 Hz, and it is a fair trade for a beat frequency that otherwise was not the one the user asked for.

2.4 Placing pvshift Within a Wider Design Space

pvshift is one point in a considerably larger space of ways to embed a fixed differential into complex audio, surveyed systematically while preparing this revision (Table 1). Pure sine carriers (**poscil**) remain cleanest when the source need not be fixed audio; a Hilbert-transform single-sideband shift avoids the phase vocoder’s windowing and FFT latency entirely, at the cost of a more involved signal chain; and spatial or amplitude-domain techniques (**vdelay3** ITD modulation, **clfilt** notch filtering, **pan2** panning) sidestep spectral processing altogether, which matters for a browser-deployed successor where FFT-heavy opcodes are harder to guarantee across platforms. We have not implemented all twelve in the current plugin; the table is the candidate set for Section 4’s successor.

3 Application Design and User Feedback

3.1 Operational Modes

The plugin offers three operational modes:

Table 1. Complex-audio entrainment techniques surveyed for this revision. **pvscale** is the superseded v1 mechanism; **pvshift** is used here.

Technique	Csound mechanism	Note
Pure-tone carrier	poscil	Cleanest signal; needs a synthesized tone, not source audio
Spectral shift	pvshift	Fixed Hz offset on every bin (used in this revision)
Spectral scale	pvscale	Ratio-based; beat drifts with partial frequency (v1, superseded)
Time-domain SSB	hilbert + quad. poscil	Zero FFT latency, exact Hz offset
Multiband bin shift	pvsbin	Shifts chosen bins only; protects bass/treble coherence
Notch filtering	clfilt	Zero pitch or added-tone artifacts
Isochronic panning	pan2, LFO	Works over speakers, no headphones needed
Interaural time delay	vdelay3	Models the physical interaural arrival-time cue
Comb-filter sweep	flanger	Organic “breathing” spatial effect
Granular pitch offset	sndwarp	Preserves transients better than phase-vocoder methods
Pitch-tracking carrier	pitchamdf + poscil	Sub-bass carrier follows the source’s key
Psychoacoustic masking	pinker, pvmaska	Hides the carrier under a dense mix

1. **Pure Sine Wave (Clinical Baseline).** Generates high-precision raw acoustic stimuli via **poscil** as an unmodulated control condition, useful for direct comparison with entrainment-enabled material.
2. **Sample Player.** Allows users to stream any familiar music track; the phase vocoder embeds the theta differential while preserving melody, harmony, timbre, and pitch. The feedback below was gathered on the v1, **pvscale**-based signal path.
3. **Generative Synthesis.** Features algorithmic compositions, currently including two ambient soundscapes by Iain McCurdy [10,6], which mitigate auditory fatigue and support sessions exceeding 30 minutes.

Users can save named presets (carrier frequency, beat frequency, texture, and spatial parameters) for quick recall across sessions.

3.2 Qualitative Case Studies

A small cohort of early adopters provided feedback over a two-week period. Younger users reported that embedding theta differentials into familiar tracks removed the emotional friction of clinical sine waves; students reached for Sample Player during focused study, while those seeking down-regulation gravitated toward Generative Synthesis [7]. Subjective sleep latency, the project’s original motivation, dropped over several weeks from over an hour to roughly twenty minutes [5,4]. The most consistent feedback, across both study and sleep use, concerned the soundstage rather than the entrainment itself: binauralized versions were almost universally described as more immersive and three-dimensional.

That observation is worth reading against Section 2.3: some of the perceived “spaciousness” may be the very spectral artifact the reviewer flagged, an inharmonic halo around the original partials, rather than a purely spatial effect. Whether **binauralizeShift** still produces that impression is an open question we intend to answer with the same cohort before the successor’s release.

4 Future Work

Binaural Explorer: Theta is deliberately narrow: one plugin, one mechanism, one target band, built to run this exploration on real listeners rather than only on paper. The **pvscale**-to-**pvshift** correction in Section 2.3 is a product of that narrowness, a problem we could only find by shipping something and having a colleague push on it; we are treating that as the model going forward, not an exception to correct for.

The next step is a considerably more capable Binaural Explorer that implements several rows of Table 1 at once and lets a user A/B them against the same source in real time, rather than committing to one spectral method per instance. Unlike the current plugin, it is planned as a permanent Csound Brain Lab module rather than a one-off class project, and will also exist as a standalone web application, open to outside researchers without installing Cabbage or Csound. Because mechanisms like **pvshift** and **vdelay3** ITD modulation generalize to any target differential, its frequency control need not be hardcoded to theta: sibling configurations will cover delta (0.5–3 Hz), alpha (8–12 Hz), beta (13–30 Hz), and gamma (30–80 Hz) within one interface. A further direction is a real-time EEG channel that would let researchers weigh Table 1’s techniques against pre- and post-session spectral data, rather than subjective report alone.

Conclusion

Binaural Explorer: Theta is an open-source first step toward using Csound and Cabbage for real-time auditory entrainment. Publishing both its original **pvscale** flaw and the **pvshift** fix, alongside the wider technique survey in Table 1, is meant to make it a useful worked example as much as a working plugin. The multi-technique successor described above is where that idea goes next.

References

1. Boulanger, R. (ed.): The Csound Book: Perspectives in Software Synthesis, Sound Design, Signal Processing, and Programming. MIT Press, Cambridge (2000)
2. Walsh, R.: Cabbage: A Framework for Audio Plugin Development. In: Proceedings of the International Csound Conference, Hanover (2011)
3. Lazzarini, V.: Spectral Music Design: A Guide to Coding Generation and Manipulation with Csound. Oxford University Press, Oxford (2014)
4. Gillan, J., Zeidan, M.S., Harris, J.M.S.: Evaluating the Efficacy of Headspace on Stress and Mindfulness: A Randomized Controlled Trial. *Scientific Reports* 11(1), 14321 (2021)
5. Huberty, J., Green, J., Glissmann, C.: Efficacy of the Calm Application on Sleep Disturbance in Adults: A Randomized Controlled Trial. *PLOS ONE* 14(12), e0225748 (2019)
6. Korotkov, N.: Procedural Sound Generation and Algorithmic Masking in Adaptive Soundscapes. *Journal of the Audio Engineering Society* 69(4), 244–252 (2021)
7. Bavafa, A., Foroughi, A., Jaberghaderi, N., Khazaei, H.: Investigating the Efficacy of Theta Binaural Beat on the Absolute Power of Theta Activity in Primary Insomniacs. *Basic and Clinical Neuroscience* 14(3), 331–340 (2023)
8. Huang, T.-L., Charyton, C.: A Comprehensive Review of the Psychological Effects of Brainwave Entrainment. *Journal of Alternative and Complementary Medicine* 14(11), 1–9 (2008)
9. Seidel, D.: Mystery Bear Music. <https://mysterybear.net/>
10. McCurdy, I.: Ambient and Generative Audio Works. <http://www.iaimccurdy.org/>
11. The Csound FLOSS Manual / Canonical Csound Reference Manual: entries for `pvshift`, `pvscale`, `pvsbin`, `hilbert`, and `vdelay3`. <https://csound.com/docs/manual/>